

PAP521S - PROGRAMMING IN PRACTICE
PROJECT A: MUNICIPAL FINANCIAL MANAGEMENT SYSTEM
INDIVIDUAL CONTRIBUTION RECORD

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Assigned Responsibility: Role 5 - Validation, Testing, Documentation & Git

1. Overview

I was responsible for validation, testing, documentation and Git support for the Municipal Financial Management System (MFMS). My work focused on making the program more reliable, checking that the modules work together correctly, and keeping the project repository clear and ready for submission.

2. Files Created / Updated

- validation.c - contains shared input validation functions used by the system.
- validation.h - declares the validation functions used by the other source files.
- tests/test_system.py - contains automated tests for the complete system.
- README.md - contains project setup, compilation and usage instructions.
- CONTRIBUTION.md - records the group members and their assigned work.

3. Tasks Completed

- Implemented shared validation for menu choices, numbers and text input.
- Applied and checked validation in main.c, employees.c, budget.c, suppliers.c, assets.c and reports.c.
- Tested normal input as well as invalid input, capacity limits and end-of-file handling.
- Documented the source code, project structure and instructions for running the program.
- Organised the Git/GitHub project files and contribution records for submission.

4. Integration with Other Modules

- The validation functions are shared by the different modules so that input is handled consistently.
- I checked that the modules compile and run together through the main program.
- I checked the repository structure and documentation so the complete project can be reviewed and run correctly.

5. Testing

- Ran the automated test suite and confirmed that all 12 tests passed.
- Checked reports, budgets, suppliers and assets using valid test data.
- Checked invalid input, overflow/capacity cases and end-of-file handling.
- Verified that the complete system starts, accepts input and produces the expected results without breaking between modules.

6. Conclusion

My contribution improved the reliability and submission quality of the MFMS by adding shared validation, testing the complete system, documenting the project and keeping the Git/GitHub repository organised.